

Ashraful Islam

AI Engineer | Machine Learning & DataOps Intern | Generative AI & Agentic Systems

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PROFESSIONAL SUMMARY

Aspiring **AI Engineer** and CSE undergraduate at CUET with research publications in **RAG, Agentic AI, and Computer Vision**, and industry experience in **Machine Learning & DataOps at Poridhi.io**. Proficient in **Python, Generative AI, Agentic AI (LangChain, LangGraph), LLM Fine-Tuning, and MLOps (Docker, FastAPI, CI/CD)**. Proven track record of architecting scalable intelligent data pipelines, optimizing inference endpoints, and delivering production-ready AI systems.

WORK EXPERIENCE

Poridhi.io

Machine Learning & DataOps Intern

Dhaka, Bangladesh (Hybrid)

August 2026 – Present

- Engineered end-to-end automated **DataOps pipelines** for data ingestion, quality validation, and feature preprocessing, supporting scalable machine learning model training workflows.
- Developed and containerized intelligent AI microservices and RESTful inference endpoints using **Python, FastAPI, and Docker**, ensuring high availability and low latency.
- Implemented **MLOps** workflows, including experiment tracking, Dockerized reproducible runtime environments, and automated CI/CD deployment pipelines for model artifacts.
- Researched and integrated emerging **GenAI** and **Agentic AI** frameworks (LangChain, LLM APIs) for internal automation and intelligent developer tools.

EDUCATION

Chittagong University of Engineering and Technology (CUET)

B.Sc. in Computer Science and Engineering – CGPA: 3.52 / 4.00 (122 Credits Completed)

Chittagong, Bangladesh

2023 – Present

- Undergraduate Thesis:** *BondhuAI-BN: Culturally-Aware Bengali Mental Health Support via Scoped Multi-Agent RAG with Domain-Adaptive Fine-Tuning.*

Dhaka City College

Higher Secondary Certificate (HSC) – Science – GPA: 5.00 / 5.00

Dhaka, Bangladesh

2019 – 2021

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, JavaScript, SQL, C#
- Generative AI & LLMs:** Agentic AI, LangChain, LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Fine-Tuning (LoRA), Vector DBs (FAISS, Chroma), Hugging Face, Gemini API
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Computer Vision (Vision Transformers/DeiT), NLP, Reinforcement Learning (DQN), Explainable AI (SHAP)
- MLOps & DataOps:** Docker, FastAPI, MLflow, CI/CD Pipelines, Data Engineering, Optuna, Model Deployment, Linux, Git/GitHub
- Backend & Databases:** Django, Django REST Framework, PostgreSQL, MySQL, SQLite, MongoDB, RESTful APIs, React

FEATURED AI / ML PROJECTS

BondhuAI-BN: Scoped Multi-Agent RAG & Fine-Tuning

🐙 GitHub

Tech: Python, LangChain, LangGraph, LoRA, Hugging Face, PyTorch, FastAPI

- Architected a culturally-aware Bengali mental health counseling support system utilizing an **Agentic AI** multi-agent framework for intent routing, empathetic response synthesis, and safety guardrails.
- Implemented domain-adaptive fine-tuning (LoRA) on open-source LLMs and integrated scoped vector retrieval over domain corpora to minimize hallucinations and enhance contextual relevance.

CCMK-RAG: Content-Conditioned Watermarked RAG

🐙 GitHub

Tech: Python, PyTorch, Transformers, FAISS, LangChain, RAG Pipelines

- Developed a passage-level attribution framework using content-conditioned multi-key watermarking in Retrieval-Augmented Generation to verify source text provenance in LLM outputs.
- Achieved high watermark recovery and attribution precision across benchmark document collections; accepted for presentation at **IEEE OMLET 2026**.

FEATURED AI / ML & ENGINEERING PROJECTS (CONTINUED)

Personal AI Assistant: Agentic Workflows & Automation



Tech: Gemini API, LangChain, Streamlit, OOP Architecture, Vector Search, Tool Calling

- Built an interactive AI assistant featuring multi-turn conversational memory, external tool integration, and document search leveraging Gemini API and LangChain.
- Designed with extensible OOP patterns, enabling automated task execution, query routing, and real-time streaming responses in a clean Streamlit interface.

Intelligent Traffic Light Control via Reinforcement Learning



Tech: Python, Deep Q-Networks (DQN), SUMO Simulator, PyTorch/TensorFlow, NumPy

- Formulated an adaptive traffic signal control system using Deep Q-Networks (DQN) integrated with the SUMO simulator to dynamically adjust phase duration based on queue density.
- Reduced average vehicle wait times and urban intersection congestion by over 25% compared to static signal timing baselines.

E-Commerce Fraud Detection Platform



Tech: Python, Scikit-Learn, XGBoost, SMOTE-Tomek, Optuna, Flask, Docker

- Developed an end-to-end fraud classification engine handling extreme tabular class imbalance via SMOTE-Tomek resampling and Optuna hyperparameter optimization.
- Packaged model pipelines into a containerized Flask REST API delivering sub-100ms real-time transaction scoring.

ResQNet: Real-Time Disaster Resource Coordination Platform



Tech: Django REST Framework, React, PostgreSQL, Docker, Concurrency Control

- Engineered a disaster resource coordination platform with an auditable need-lifecycle (pending → open → committed → resolved) and atomic conditional updates to prevent race conditions.
- Enforced server-side role-based access control (RBAC) and packaged services with Docker Compose for production deployment.

RESEARCH PUBLICATIONS

- **CCMK-RAG: Content-Conditioned Multi-Key Watermarking for Passage-Level Attribution in Retrieval-Augmented Generation**
Towhidul Islam, Ashraful Islam, Mahfuzul Hoq Chowdhury – *IEEE OMLET 2026 (Accepted)*.
- **Section-Aware Retrieval-Augmented Generation for Bangladesh Labour Law Question Answering: A Comparative Study**
Towhidul Islam, Ashraf-ul-Islam, Mahfuzulhoq Chowdhury – *IEEE OMLET 2026 (Accepted)*.
- **An Explainable Machine Learning and Ensemble Voting Classifier for Cyber Attack Detection in Highly Imbalanced IoT Networks**
Ashraful Islam, Towhidul Islam, Dr. Mahfuzulhoq Chowdhury – *IEEE 2026 3rd International Conference on Data Science and Business Systems (ICDSBS 2026) (Accepted)*.
- **Vision Transformer-Based Detection of Counterfeit Bangladeshi Banknotes Using the DeiT-Tiny Architecture**
Ashraful Islam, Dr. Mahfuzulhoq Chowdhury, Towhidul Islam – *International Conference on Circuit Power & Computing Technologies (ICCPCT-2026) (Accepted)*.
- **A Double Layer Stacking Classifier Based Explainable Tax Evasion Fraud Prediction with Optuna and SMOTE-Tomek**
Dr. Mahfuzulhoq Chowdhury, Saklain Ahmed, Ashraful Islam – *1st International Conference on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONNECT 2026) (Submitted)*.
- **A Traffic Rules Violation Prediction Scheme Based on Hybrid Stacking Ensemble Classifier and SHAP Analysis**
Dr. Mahfuzul Hoq Chowdhury, Oarisa Rebayet, Ifath Jeba Chowdhury, Ashraf-ul-Islam, Isnat Mehrin Sami, Snaholata Mondal – *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) (Copyright Confirmed)*.

CERTIFICATIONS & TRAINING

- **Full Stack Data Science with Generative AI Bootcamp 1.0** – Inception BD
- **Complete Data Science, Machine Learning, DL, NLP Bootcamp** – Udemy (October 2025)
- **SQL (Intermediate) & SQL (Basic)** – HackerRank (December 2024, Valid to 2034)
- **Database Structures and Management with MySQL** – Meta Professional Certificate
- **Exploratory Data Analysis in Python** – Udemy (February 2026)

COMPETITIVE PROGRAMMING & LEADERSHIP

- **Competitive Programming:** Solved **500+ problems** across platforms – Codeforces (*The_phoenix*, Max Rating: 1133, 335+ solved), LeetCode (*noobashraf21*, 120+ solved), and HackerRank.
- **Leadership & Extracurriculars:** **General Secretary** – Cox's Bazar Student Forum, CUET; **Coordinator** – Cox's Bazar Student Society. Organized tech seminars, student welfare initiatives, and academic mentoring.